

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			Substitution: $3 \times 10^8 = \frac{\text{distance}}{870 \times 31536\,000}$ (1) Rearrangement: distance = $3 \times 10^8 \times 870 \times 31\,536\,000$ (1) Distance = 8.23×10^{18} [m] (1) Award 2 marks for answer of 8.23×10^n [m] where n isn't 18 Award 2 marks for answer of 2.61×10^{11} [m] Award 2 marks for answer of 9.46×10^{15} [m] Award 1 mark for answer of 2.61×10^n [m] where n isn't 11 Award 1 mark for answer of 9.46×10^n [m] where n isn't 15 Award 1 mark for answer of 2.74×10^{10} [m]	1	1 1		3	3	
	(b)	(i)		Supergiant (1) Supernova (1) Neutron star or black hole (1)	3			3		
		(ii)		<u>Heavy</u> elements {released / returned to space} (1) during supernova (1)	2			2		
		(iii)		Cloud of gas and dust (1) collapsed due to gravity OWTTE (1)	2			2		
				Question 3 total	8	2	0	10	3	0